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Effect of pressure and fuel–air unmixedness on NO_x emissions from industrial gas turbine burners

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Abstract

The effect of fuel–air unmixedness on NO_x emissions from industrial lean premixed gas turbine burners fueled with natural gas is analyzed in the pressure range from 1 to 30 bar. The analysis is based on a model where NO_x production is split, according to a Damköhler number criterion, into a "prompt" (fast) contribution generated within the very narrow instantaneous heat release region (flamelet) and a "postflame" (slow) one, generated in the combustion products. Using GRI3.0 chemical kinetics, it is found that (a) the prompt NO_x contribution is approximately a factor of 3 less sensitive to adiabatic flame temperature variations than postflame NO_x and (b) prompt and postflame NO_x change with pressure respectively according to an exponent $\alpha_{PR} \simeq -0.45$ and $\alpha_{PF} \simeq 0.67$. It is shown that total NO_x emissions change from being mostly of prompt type at 1 bar to being mostly of postflame type at 30 bar, so that the effect of fuel–air unmixedness on NO_x emissions significantly increases with increasing pressure. The combination of these findings yields a negative NO_x pressure exponent under fully premixed conditions across a rather large range of equivalence ratios but a positive one for levels of fuel–air unmixedness typical of industrial burners. This result is confirmed by the application of the NO_x model in the large eddy simulation of the ALSTOM EV double cone burner, which gives, in line with experimental data, an NO_x pressure exponent growing, with equivalence ratio, from $\simeq$0.1 to $\simeq$0.67.
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Keywords: Turbulent partially premixed combustion; NO_x formation; Large eddy simulation

1. Introduction

Developing lean premixed industrial gas turbine burners is a complex and expensive task. Additionally to high pressure testing, in order to keep development costs to a reasonable level, gas turbine manufacturers also make significant use of atmospheric tests. The direct extension of atmospheric data to the high-pressure range (15–30 bar) in gas turbine combustors fueled with natural gas (NG) seems reasonable for all those aspects related to flame position. NG is in fact mostly composed of methane, a fuel component that, when premixed with air at the high Damköhler and Reynolds numbers usually found in gas turbine combustors, gives flames whose turbulent burning rate U_t does not strongly depend on the operating pressure p. For example, Bunsen burner experiments reported in [1] have shown a $U_t \simeq p^0$ dependency.

The situation is less clear for nitric oxides (NO_x) emissions, whose formation has in the past been the

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Nomenclature

A	constant in TFC model
A_0	mean flame cross-sectional area
c	reaction progress variable
D	molecular diffusion coefficient
$\mathrm{Da} = \tau_\mathrm{m}/(\lambda/s_\mathrm{L}^2)$	Damköhler number
D_t	turbulent diffusion coefficient
$D_\mathrm{t,F}$	turbulent diffusion coefficient within flamelet thickness
f	mixture fraction
l_K	characteristic length scale across turbulent spectrum
l_t	integral length scale of turbulence
n_PR	prompt NO_x [kmol/kg]
$n^\star_\mathrm{PR}(f)$	prompt NO_x from CHEMKIN–PREMIX
$n^{(\mathrm{d})}_\mathrm{PR}$	prompt NO_x correction
$n^{(\mathrm{ref})}_\mathrm{PR}(f)$	reference prompt NO_x
p	pressure
p_0	reference pressure
$P(f)$	mixture fraction probability density function
s_L	laminar flame speed
S	NO_x sensitivity to temperature
t	time
T	temperature of combustion products
T_a	activation temperature used in the Arrhenius type approximation of the NO_x concentration
T_reac	temperature of reactants
$T_\mathrm{REF} = 1800$ K	reference temperature used for nondimensionalization
u_K	characteristic velocity across turbulent spectrum
u'	turbulent velocity fluctuation
u'_F	turbulent velocity fluctuation within thickened flamelet
U	fuel–air unmixedness factor
U_t	turbulent burning rate
w_PR	prompt NO_x source
$w^{(\mathrm{d})}_\mathrm{PR}$	source of prompt NO_x correction
w_PF	source of postflame NO_x
X	nondimensional concentration
X_0	NO_x concentration at reference conditions
$\mathbf{X}$	coordinate on flamelet surface
Y_j	mass fraction of species j

Greek letters

α	NO_x pressure exponent
δ_F	size of thickened flamelet
ϵ_f	mixture fraction scalar dissipation rate
λ	molecular conductivity
ξ	mixture fraction in composition space
ρ	density
τ_m	turbulent mixing time
$\tau_\mathrm{HR} = \lambda/s_\mathrm{L}^2$	heat release time scale

Symbols

$(\,)_\mathrm{b}$	value in products stream
$(\,)_\mathrm{u}$	value in reactants stream
$\overline{(\,)}$	Reynolds averaging
$\widetilde{(\,)}$	Favre averaging
$(\,)'' = (\,) - \widetilde{(\,)}$	Favre fluctuation

subject of several experimental and analytical studies [2–10]. A very common assumption for NO_x pressure scaling is the $\simeq p^{0.5}$ dependency [2]. In this case, extrapolation from 1 bar to high pressure would be possible on the basis of just a universal pressure scaling exponent. Other investigators have, however, reported a more complicated situation, sometimes even with a negative pressure exponent dependency (NO_x decreasing with pressure). For example, a reduction of NO_x emissions with pressure has been observed in the experiments performed in [3–5]. The experiments in a high-pressure jet-stirred reactor performed in [3] have shown that, at a combustion temperature of 1800 K and a residence time around 2 ms, NO_x emissions decrease with pressure and this reduction is more pronounced in the range up to 5 bar than in the range above it. Similar NO_x behavior was observed in the experiments performed in [4,5] for the same kind of burner, with [5] reporting a negative pressure scaling exponent equal to -0.2. In a review paper [6], it was suggested that NO_x emissions scale with pressure ($[NO_x]/[NO_x]_{1\ \mathrm{bar}} = p^\alpha$) with an exponent α that increases from $\alpha = 0$ under very lean premixed conditions to $\alpha = 0.5$ for higher equivalence ratios. It was instead concluded in [7], on the basis of experiments, that NO_x emissions in well premixed flames are independent of pressure and suggested that, in well-designed lean premixed burners, it is possible to operate a combustor from 1 to 30 bar without any significant change in NO_x emissions. In these experiments the pressure dependency was found to increase only slightly with increasing equivalence ratio. This conclusion was also reached in [8], where experiments indicated that NO_x emissions are independent of pressure at an equivalence ratio of 0.5 but that pressure dependency increases as soon as the fuel–air unmixedness level is increased. Given these results from previous experimental investigations, the primary scope of the present work is to shed additional light on the combined effect of pres-

sure and unmixedness on NO_x emissions, especially from industrial gas turbine burners.

Turbulent combustion in industrial lean premixed burners takes place at high Reynolds and, for fuels like natural gas, moderately high Damköhler (Da) numbers. Under these conditions, heat is released in regions that are much smaller than the integral length scale of turbulence; i.e., combustion takes place in the so-called flamelet regime. However, typical values of the Karlovitz (Ka) number, i.e., the ratio between the heat release and the Kolmogorov time scales, are usually still larger than unity, meaning that the smallest turbulent eddies can penetrate the preheating region, which therefore becomes subject to an increase of thickness until a new equilibrium between turbulent internal diffusion and heat release is reached. The problem of turbulent premixed combustion under Da > 1, Ka > 1 conditions was first investigated in [11] where the final size δ_F of the flamelet broadened by small-scale turbulence (from which comes the name of *thickened flamelet* combustion regime) was estimated from the turbulent diffusion law as $\delta_F = \sqrt{D_{t,F}\tau_{HR}}$, where $D_{t,F}$ is the diffusion coefficient due to vortices of size equal and smaller to the flamelet thickness δ_F, i.e., $D_{t,F} \simeq u'_F\delta_F$. The turbulence intensity associated with these vortices can be estimated from the equilibrium of turbulence kinetic energy flux across the turbulent spectrum (i.e., $u'^3/l_t = u'^3_k/l_k$) as $u'_F \simeq u'(\delta_F/l_t)^{1/3}$, which, used within the expression for δ_F, gives, after some algebraic manipulation, the final value of $\delta_F \simeq l_t \mathrm{Da}^{-3/2}$. This expression shows that the thickened flamelet has a size that is a $\mathrm{Da}^{-3/2}$ fraction of the integral length scale of turbulence. In the case of Da > 1, small-scale turbulence with size larger than δ_F has the effect of wrinkling the flamelet, increasing its area and therefore the mean total burning rate.

For consistency with the above flamelet structure of the preheating and heat release regions, NO_x formation is split in this work into two contributions according to a Damköhler number criterion: a fast one (called here "prompt"), which is the NO_x part formed within the flamelet region, and a slow-forming part (called here "postflame"), which takes place in the hot gases downstream of the instantaneous heat release region. A detailed study of the turbulence–chemistry interaction effects inside the thickened flamelet structure would be too demanding for the scope of the present investigation. The analysis has therefore been carried out assuming an instantaneous turbulent premixed flamelet with a structure like that of a laminar freely propagating flame. This assumption is partially justified by the fact that combustion reactions are characterized by high activation energy. In this case the thickness of the heat-release region is much smaller than the thickness of the preheating one, this last, for Da > 1, Ka > 1, broadened by small-scale turbulence. It can therefore be reasonably assumed that the instantaneous-heat-release region, where prompt NO_x formation takes place, has a size smaller than that of the Kolmogorov vortices and therefore remains laminar. The strain effect produced by the Kolmogorov vortices on the laminar heat release region is also neglected. This could eventually be accounted for replacing the freely propagating flame structure with one of counterflowing laminar flames where the strain rate is an independent parameter to match to the Kolmogorov strain rate.

The present work is organized as follows. In Section 2, NO_x formation is analyzed under typical conditions of gas turbine combustors and in the case of fully premixed reactants, with the CHEMKIN II–PREMIX module [12] and GRI3.0 [13] chemical kinetics. The analysis is supported via experimental data in a fully premixed, strongly adiabatic dump combustor that was already used in previous experimental investigations of turbulent premixed flames [14].

In Section 3, a computational model to account for the combined effect of unmixedness and pressure on NO_x emissions is developed. The model is developed for CFD simulations of industrial low-NO_x burners. In this section the model is used according to a simple 1D formulation in order to obtain a preliminary idea of the effect of unmixedness and pressure on NO_x emissions from industrial gas turbine burners.

In Section 4, the NO_x model is used in a large eddy simulation of the combusting flow generated by a prototype version of the low-emissions ALSTOM EnVironmental (EV) double cone burner to confirm findings from the simple 1D analysis. CFD results are in this case validated with experimental data at 1 and 15 bar.

2. NO_x formation under perfectly premixed conditions

Fig. 1 shows results from CHEMKIN II–PREMIX [12] and GRI3.0 mechanisms [13] at 1 and 30 bar for an adiabatic temperature of combustion products within typical values found in gas turbines, i.e., 1700 K < T < 1900 K (NO_x and temperature nondimensionalized hereafter with reference values T_{REF} = 1800 K, NO_x reference concentration = 2 ppmv dry at 15% O_2). These results indicate that NO_x formation is characterized by an initial step within the narrow heat-release region and a more slowly growing contribution in the postflame region.

NO_x formation is usually studied in terms of chemical kinetics pathways. From this point of view NO_x formation is decomposed, see [15], into four

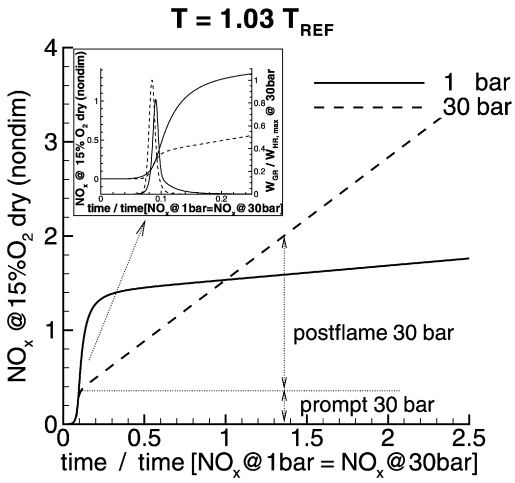

Fig. 1. NO_x distribution from the CHEMKIN II–PREMIX module, GRI3.0 chemistry versus time for 1D freely propagating NG–air premixed flame at two different pressures and a given adiabatic flame temperature.

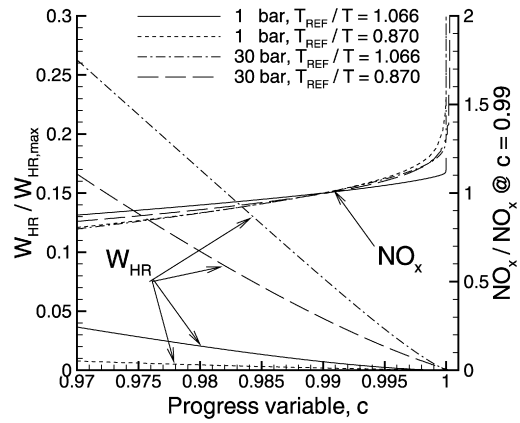

Fig. 2. Sensitivity of the prompt NO_x level to the choice of the progress variable threshold value.

contributions: (a) Zeldovich slow reaction $N_2 + O \Rightarrow NO + N$, which forms two NO molecules, one directly and the second indirectly from the fast oxidation of the N atom via the second Zeldovich reactions $N + O_2 \Rightarrow NO + O$, (b) N_2O, where the N_2O formed by the three-body reaction $N_2 + O + M \Rightarrow N_2O + M$ is attacked by the O-atom to give two molecules of NO and by the H-atom, leading to a molecule of NO and a molecule of NH, which is then rapidly oxidized to NO, (c) the Fenimore contribution, where, within the flame zone, N_2 is attacked by the CH radical, $N_2 + CH \Rightarrow HCN + N$, and the HCN, N radicals are rapidly converted to NO, (d) NNH pathway, where the reaction $NNH + O \Rightarrow NO + NH$ gives two molecules of NO (one directly and one indirectly via fast oxidation of the NH radical).

In the present study, a different point of view is adopted. NO_x formation is split here into two contributions according to a Damköhler number criterion: a fast one (called here "prompt") that is the NO_x part formed within the heat-release region and a slow-forming part (called here "postflame") that takes place in the hot gases downstream of the instantaneous-heat-release region. This decomposition of NO_x formation is dictated by the idea that turbulence–chemistry interaction effects must be modeled consistently with their characteristic Damköhler numbers. Each species that is (promptly) produced or consumed within the flamelet, from initial mass fraction $Y_{j,\mathrm{u}}$ to final value $Y_{j,\mathrm{b}}$, is released by the mean flame element of area $\mathrm{d}A_0$ at a rate (mass/time) given by $\rho_\mathrm{u} U_\mathrm{t}(Y_{j,\mathrm{b}} - Y_{j,\mathrm{u}})\,\mathrm{d}A_0$. This point of view leads to a modeling concept based on the turbulent burning rate (see for example [11,16]) that accounts at the same time for the mean flame and also for any species that is produced within the flamelet sheet. Quantities that are instead more slowly (volumetrically) produced in the postflame region can be accounted for via a plug flow reactor approach coupled, if fuel–air unmixedness is still present, with a model to account for turbulent fluctuations in the fuel mixture fractions.

As mentioned already in the Introduction, prompt NO_x emissions are determined assuming a flamelet sheet with structure equal to that of the freely propagating laminar flame. Prompt NO_x is therefore defined as the NO_x concentration in the laminar freely propagating flame at the location where combustion is close to equilibrium, i.e., where the progress variable c (giving the degree of conversion of the fuel–air mixture into its products) is sufficiently large so that heat release has almost completed. A prompt NO_x cutoff value for the progress variable equal to unity would, however, be inadequate because it includes also the full region of combustion products, where also the postflame NO_x contribution is produced. A suitable value of the progress variable to detect the level of prompt NO_x has been determined here from the plot of NO_x concentration across the laminar flame versus the reaction progress variable shown in Fig. 2 for two flame temperatures and two values of the operating pressure ($0.97 < c < 1.0$). The plot shows that for $c > 0.99$ the heat release is almost completed in all cases (all heat release rates are below 7% of their maximum value) and also that the prompt NO_x level does not change by more than 10–20% of the value at $c = 0.99$, which is the threshold value used in this work. Values higher than this threshold one can also be used (but not $c = 1$), but this would not alter the final conclusions significantly.

Postflame NO_x is instead estimated from the formation rate in the products region. In general, at flame temperatures and residence times that are typical of

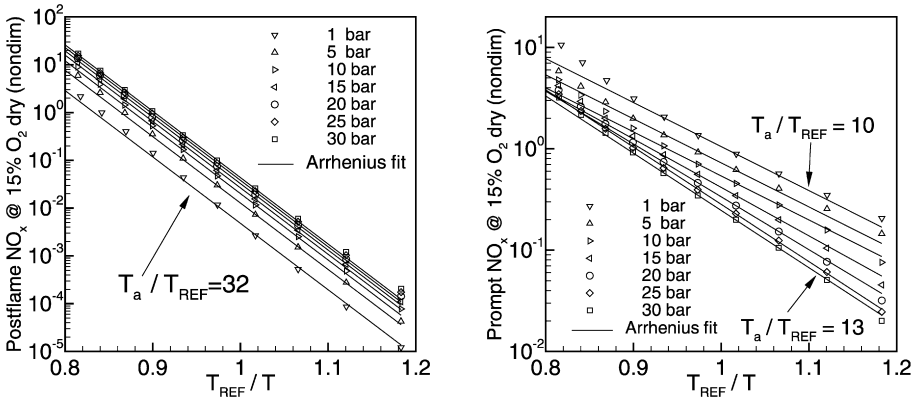

Fig. 3. Postflame (left) and prompt (right) NO_x from CHEMKIN II, GRI3.0 mechanism versus adiabatic flame temperature for NG/air premixed mixture.

gas turbine burners, this formation rate is constant (Fig. 1); i.e., there is a negligible effect of backward reactions, which cause the total NO_x production rate to decrease to zero.

Prompt NO_x and postflame NO_x concentrations are plotted here as functions of the adiabatic temperature of the combustion products T at different values of the operating pressure. It has been preferred to plot the NO_x concentrations versus the temperature T of the combustion products because of the direct link between this variable and NO_x formation rates via the exponential Arrhenius term $\simeq \exp(-T_a/T)$. The equivalence ratio ϕ does not instead link unequivocally to the adiabatic flame temperature because of the offset given by the initial temperature of the reactants. All experimental data shown in this work refer to fuel-lean conditions ($\phi < 1$) and a nondimensional initial temperature of reactants $T_{react}/T_{REF} = 0.4$. Results from CHEMKIN are given as symbols in Fig. 3 in logarithmic scale versus T_{REF}/T (postflame NO_x given for 1 ms residence time).

Approximating these data with the Arrhenius-type expression (shown with lines in Fig. 3)

$$X_{NO_x} = X_0 \left(\frac{p}{p_0}\right)^{\alpha} \exp\left[-T_a/T_{REF}(T_{REF}/T - 1)\right], \tag{1}$$

where T_a has the role of activation temperature, yields the coefficients given in Table 1 ($p_0 = 15$ bar; X_0 is the NO_x concentration at p_0 and T_{REF}). Fig. 4 shows instead the approximation of the postflame and prompt NO_x at $T = T_{REF}$ with the pressure scaling $(p/p_0)^{\alpha}$.

These NO_x trends lead to the following conclusions:

- Within the range of flame temperatures in which gas turbine burners operate, nondimensional activation temperatures T_a/T_{REF} used in the approximation (1) of prompt and postflame NO_x are respectively equal to $\simeq$10–13 and 32 and rather independent of pressure (a mild increase with pressure is observed for the prompt NO_x above 10 bar). Postflame NO_x is therefore characterized by a sensitivity to flame temperature variations (and therefore unmixedness) that is much higher than the sensitivity of prompt NO_x.
- While postflame NO_x increases with pressure according to the pressure exponent $\alpha \simeq 0.67$,

Table 1
Parameters for the approximation of prompt and postflame NO_x distributions given by relation (1) ($p_0 = 15$ bar)

	Postflame	Prompt
X_0	0.0264	0.42
α	0.67	−0.45
T_a/T_{REF}	32	10–13

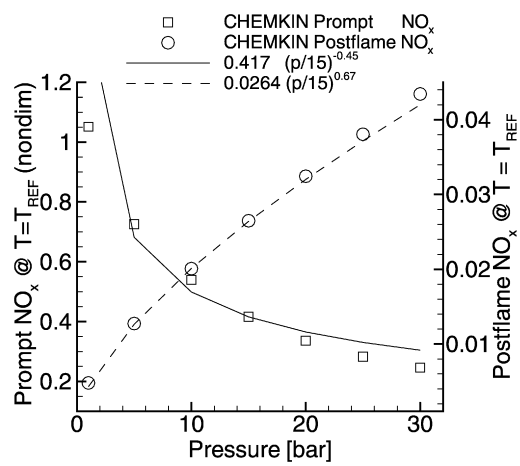

Fig. 4. Postflame and prompt NO_x variation with pressure at $T = T_{REF}$.

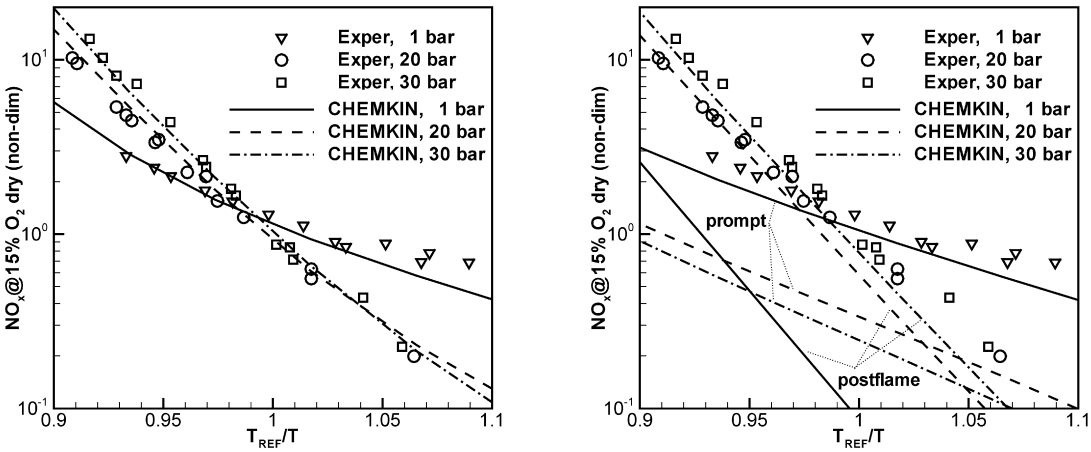

Fig. 5. Comparison of NO_x from CHEMKIN II–PREMIX with experiments under complete fuel–air premixing for NG/air. Left: total NO_x. Right: postflame and prompt NO_x individual contributions.

prompt NO_x decreases according to the pressure exponent $\alpha \simeq -0.45$.

Introducing the following parameter S, which gives the sensitivity of NO_x emissions to temperature,

$$S = \frac{1}{X_{NO_x}} \frac{dX_{NO_x}}{d(T/T_{REF})}, \tag{2}$$

we have that

$$\frac{S_{PF}}{S_{PR}} = \frac{d\ln(X_{NO_x,PF})}{d(T/T_{REF})} \Big/ \frac{d\ln(X_{NO_x,PR})}{d(T/T_{REF})} \simeq 2.5\text{–}3; \tag{3}$$

i.e., postflame NO_x emissions are approximately three times more sensitive to unmixedness than prompt NO_x.

The NO_x emissions estimated via the CHEMKIN II–PREMIX module are compared here with experimental data from burners where reactants are completely premixed prior to combustion (zero unmixedness at the flame). Three pressure levels have been considered: 1, 20, and 30 bar. The 1-bar experimental data are from a fully premixed ALSTOM EV double-cone burner that has in the past been the subject of several experimental [17] and analytical [18–20] investigations, while the 20- and 30-bar are from a fully premixed, strongly adiabatic dump combustor with a characteristic residence time of ≃20 ms. This dump combustor, available to the ALSTOM Combustion development, is based on a modular design consisting of an electric heating unit, a mixing tube that ends in a sudden area expansion in a cylindrical combustor, and a cooling/exhaust pipe. Three 30-kW electric heaters preheat the air stream to the desired temperature (0–650 °C). A 30-cm-long series of static mixers allow high levels of fuel/air premixing before the reactants reach the dump combustor. The mixing tube and the combustor are lined with ceramic insulation to promote near-adiabatic operation. Tests can be run with a maximum mass flow rate of 250 g/s. The maximum achievable pressure is 35 bar. Full details of this experimental setup can be found in [14].

It must be noticed that, under perfect fuel–air premixing and adiabatic conditions, the only relevant parameter in the formation of postflame NO_x is the product residence time. The combustor, in fact, acts as a plug flow reactor where the reacting volume V is bounded by the flame on one side and the combustor exit on the other. Products flow through this volume with a volumetric flow rate F such that the final residence time is V/F. The volume where prompt NO_x formation (and heat release) takes place is considered here small with respect to the hot gas postflame volume, consistent with the hypothesis of flamelet combustion. This hypothesis loses validity in the very-low-flame-temperature range, which, however, is not of large interest for gas turbines.

The comparison between calculated and measured NO_x emissions is shown in Fig. 5. Postflame NO_x is calculated for a residence time of 18 ms, approximately the residence time in the combustors. The right part of the figure shows a comparison of experimental NO_x emissions with the prompt and postflame contributions from CHEMKIN–PREMIX at the three pressures versus T_{REF}/T. These results clearly indicates that, in the intermediate and high-temperature range, approximately for $T_{REF}/T < 1$, high pressure (20- and 30-bar) experimental NO_x are well approximated just by the calculated postflame contribution. On the other hand, the prompt contribution alone is sufficient to give a reasonable estimate of the 1-bar NO_x emissions (slightly underestimated), especially in the intermediate and low-temperature range, approx for $T_{REF}/T > 1$.

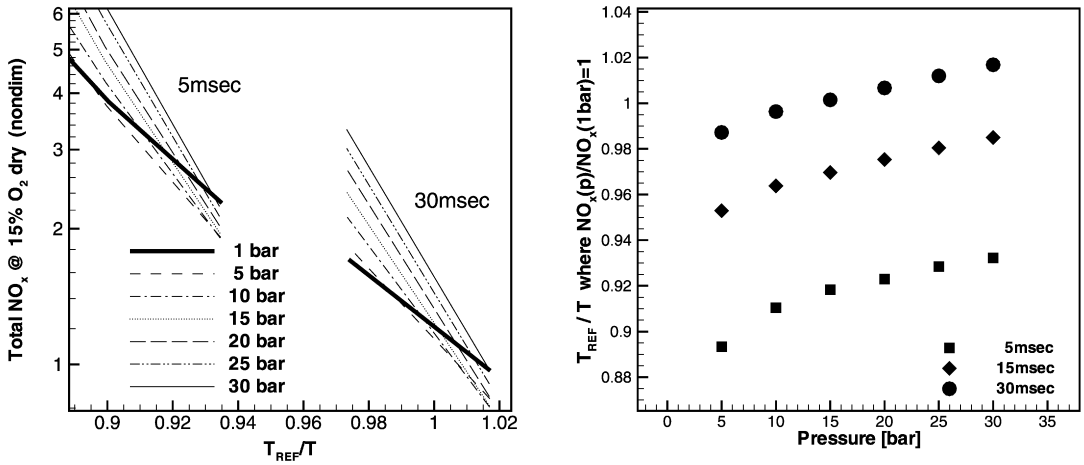

Fig. 6. Left: total NO_x emissions from CHEMKIN–PREMIX under full NG/air premixing versus adiabatic flame temperature at different pressures and residence times. Right: temperature where NO_x at given pressure p is equal to 1-bar NO_x.

At high pressure and in the low-temperature range, a better estimate of the experimental NO_x emissions is obtained by adding the prompt contribution (see Fig. 5, left). At low temperature and high pressure, in fact, both prompt and postflame NO_x are small and comparable. The situation is reversed for the 1-bar case, where postflame NO_x become comparable to prompt NO_x only in the high-temperature range and then, accounting for both of them, helps to improve here the agreement with experimental data.

When added together, prompt and postflame NO_x yield a total NO_x pressure exponent that goes from a negative value of $\simeq -0.4$ at $T_{REF}/T = 1.07$ to a positive value of $\simeq 0.6$ at $T_{REF}/T = 0.94$. In the low-temperature range, in fact, postflame plus prompt NO_x at high pressure is much lower than the prompt NO_x contribution dominating total NO_x emissions at 1 bar. Postflame NO_x, however, rises with temperature much more quickly than prompt NO_x (due to higher activation temperature), so that the situation is reversed at $T_{REF}/T \simeq 0.99$, where the high-pressure postflame contribution becomes equal to the 1-bar prompt one. Finally, at $T_{REF}/T = 0.94$, the postflame high-pressure NO_x contribution dominates above the 1-bar total NO_x, giving a positive pressure exponent of $\simeq 0.6$.

The characteristic temperature at which the 1- and 30-bar total NO_x emissions become equal decreases with increasing residence time. While, in fact, prompt NO_x does not change with residence time (it is a fixed offset), high-pressure postflame NO_x increases faster than under atmospheric conditions, leading to a reduction of the temperature at which the 1- and 30-bar NO_x are equal (see Fig. 6, left). A similar situation occurs with the increase in pressure. The flame temperature at which NO_x concentration at a pressure p becomes equal to NO_x at 1 bar (for a given value of the residence time) in fact decreases with pressure p. This result, shown in Fig. 6, is due to postflame NO_x increasing with pressure faster than the reduction in prompt NO_x, as reported in Table 1. This is also confirmed by experiments shown in Fig. 5, where the temperature at which NO_x emissions at pressure p become equal to NO_x emissions at 1 bar has a small decrease from $p = 20$ bar to $p = 30$ bar. Finally, Fig. 6 right shows the variation of the NO_x intersection temperature (temperature where $NO_x(p) = NO_x(p = 1\text{ bar})$) with pressure at different residence times.

3. Effect of unmixedness on NO_x emissions

3.1. Modeling

The experimental data presented in the previous section for the case of perfectly premixed flames indicate that the sensitivity of NO_x emissions to temperature increases with pressure. It has been shown that NO_x emissions behave in this way due to the increase in the postflame NO_x contribution and decrease of prompt (flamelet) NO_x with increasing pressure, the former much more sensitive to temperature variations than the latter.

This result suggests also that, the higher the pressure, the higher should be the effect of fuel–air unmixedness on NO_x emissions. In this section a CFD model able to account for postflame and prompt NO_x under partially premixed conditions is developed. The model is therefore applied here in a simple one-dimensional form in order to get a preliminary assessment on how fuel–air unmixedness and pressure affect NO_x emissions.

Under fully premixed conditions, the rate of prompt NO_x production for a unit of mean flame

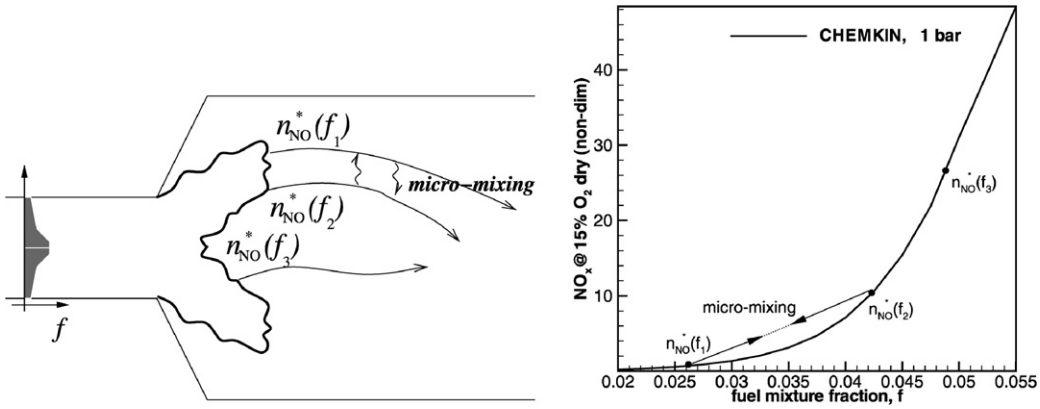

Fig. 7. Sketch of prompt NO_x release and downstream micromixing from partially premixed flamelet.

cross-sectional area can be estimated as $\rho_u U_t n^\star_{PR}$, where $n^\star_{PR}$ represents moles of NO_x per unit of mass released by the narrow flamelets. This last quantity, which is a function of the fuel mixture fraction f, is obtained here from the CHEMKIN II–PREMIX module as described in the previous section.

In industrial burners, some degree of unmixedness (partially premixed reactants) at the flame is usually necessary in order to have good flame stabilization properties. Under partially premixed conditions, the estimation of NO_x emissions becomes a more complicated issue. It is assumed here that, also in presence of partial premixing, the flamelet sheet preserves the structure of a 1D unstrained freely propagating laminar flame at the instantaneous local value of the equivalence ratio. In this case, the equivalence ratio at the flamelet sheet and thus also the concentration of released prompt NO_x are functions of time and space. This situation is accounted for with a model describing the release of prompt NO_x from the flamelet surface with concentration depending on the local instantaneous equivalence ratio and then the successive downstream micromixing of these concentrations.

The model developed here for prompt NO_x originates from the fact that molecular mixing is a linear process, while the distribution of prompt NO_x released from the flamelet (see Fig. 7, left) with concentration $n^\star_{PR}(f)$ from CHEMKIN II–PREMIX calculations is nonlinear (Fig. 7, right). The final concentration of prompt NO_x at the outlet of the combustor, where complete fuel–air premixing is achieved, is therefore different from the one obtained from a 1D premixed flame calculation at the bulk equivalence ratio in the combustor.

The effect of unmixedness on the final prompt NO_x concentration at the combustor exit is here accounted for using $n^\star_{PR}(f)$ as a boundary condition on the fluctuating instantaneous flamelet sheet for a conservation equation that describes the downstream micromixing of prompt NO_x.

Extending this idea to a Reynolds or LES (large eddy simulation) Navier–Stokes averaging framework, however, implies some difficulties. In fact, the boundary condition $n_{PR} = n^\star_{PR}[f(\mathbf{X}, t)]$, where $\mathbf{X}$ indicates points on the flamelet sheet, spreads volumetrically over the mean flame brush thickness and also involves the introduction of the probability density function for the mixture fraction f.

These difficulties are overcome by solving for a quantity $n^{(d)}_{PR}$ that gives the local deviation (called hereafter the prompt NO_x correction) of the prompt NO_x concentration n_{PR} from a reference value $n^{(ref)}_{PR}(f)$, a function only of the mixture fraction f; i.e. $n_{PR} = n^{(ref)}_{PR}(f) + n^{(d)}_{PR}$.

Replacing $n_{PR} = n^{(ref)}_{PR}(f) + n^{(d)}_{PR}$ into the instantaneous transport equation for the prompt NO_x concentration,

$$\rho \frac{\partial n_{PR}}{\partial t} + \rho \mathbf{u} \cdot \nabla n_{PR} = \nabla \cdot (\rho D \nabla n_{PR}) + \rho w_{PR}, \quad (4)$$

yields the following equation:

$$\begin{aligned} \rho \frac{\partial n^{(d)}_{PR}}{\partial t} &+ \rho \mathbf{u} \cdot \nabla n^{(d)}_{PR} \\ &= \nabla \cdot \left(\rho D \nabla n^{(d)}_{PR}\right) + \rho w_{PR} + \nabla \cdot \left[\rho D \nabla n^{(ref)}_{PR}(f)\right] \\ &\quad - \rho \frac{\partial n^{(ref)}_{PR}}{\partial t} - \rho \mathbf{u} \cdot \nabla n^{(ref)}_{PR}. \end{aligned} \quad (5)$$

Assuming $\rho D = \text{const}$ and using the algebraic transformations

$$\frac{\partial n^\star_{PR}}{\partial t} = \frac{dn^\star_{PR}}{df}\frac{\partial f}{\partial t}; \quad \nabla n^\star_{PR} = \frac{dn^\star_{PR}}{df}\nabla f;$$

$$\nabla^2 n^\star_{PR} = \frac{dn^\star_{PR}}{df}\nabla^2 f + \frac{d^2 n^\star_{PR}}{df^2}|\nabla f|^2 \quad (6)$$

plus the transport equation for the mixture fraction f,

$$\rho\frac{\partial f}{\partial t}+\rho\mathbf{u}\cdot\nabla f=\rho D\nabla^2 f, \tag{7}$$

the last three terms in (5) reduce to

$$\rho w_{\mathrm{PR}}^{(\mathrm{d})}=\rho D|\nabla f|^2\frac{\mathrm{d}^2 n_{\mathrm{PR}}^{(\mathrm{ref})}}{\mathrm{d}f^2}. \tag{8}$$

The final equation for $n_{\mathrm{PR}}^{(\mathrm{d})}$ becomes

$$\rho\frac{\partial n_{\mathrm{PR}}^{(\mathrm{d})}}{\partial t}+\rho\mathbf{u}\cdot\nabla n_{\mathrm{PR}}^{(\mathrm{d})} =\nabla\cdot\left(\rho D\nabla n_{\mathrm{PR}}^{(\mathrm{d})}\right)+\rho w_{\mathrm{PR}}+\rho D|\nabla f|^2\frac{\mathrm{d}^2 n_{\mathrm{PR}}^{(\mathrm{ref})}}{\mathrm{d}f^2}. \tag{9}$$

To specify the reference prompt NO_x distribution $n_{\mathrm{PR}}^{(\mathrm{ref})}(f)$, the computational domain is divided into three regions: (a) region of reactants, (b) region of heat release, (c) region of products. In the region of reactants, we have $n_{\mathrm{PR}}=0$. This result is obtained with (9) by using $n_{\mathrm{PR}}^{(\mathrm{ref})}(f)=0$ in the computational domain and $n_{\mathrm{PR}}^{(\mathrm{d})}=0$ on the air and fuel inlet boundaries. The same equation can be used with little error in the instantaneous flame region, i.e., the region where the progress variable c assumes values between zero and unity. In fact, the probability of finding $0<c<1$ (within the mean flame brush) is very low because the instantaneous flame is very narrow at flamelet combustion conditions. Therefore, is calculating the local mean concentration of prompt NO_x from the contributions given by the three regions, little error is made if the value of prompt NO_x within the reaction region is set to zero. In the products region, the reference prompt NO_x concentration $n_{\mathrm{PR}}^{(\mathrm{ref})}(f)$ is instead set equal to the prompt NO_x concentration $n_{\mathrm{PR}}^{\star}(f)$ obtained from CHEMKIN calculations as described in Section 2. In this case, Eq. (9) shows that the local prompt NO_x concentration in the products deviates from $n_{\mathrm{PR}}^{\star}(f)$ by an amount that depends on the nonlinearity versus f of the $n_{\mathrm{PR}}^{\star}(f)$ distribution. The boundary condition on the fluctuating flamelet surface for the prompt NO_x conservation (4) equation is then transformed in the present approach to a source term that can be more easily handled in terms of mean quantities.

Taking finally the average of Eq. (9) yields

$$\bar\rho\frac{\partial\tilde n_{\mathrm{PR}}^{(\mathrm{d})}}{\partial t}+\bar\rho\tilde{\mathbf{u}}\cdot\nabla\tilde n_{\mathrm{PR}}^{(\mathrm{d})}=\nabla\cdot\left[\bar\rho D_{\mathrm{t}}\nabla\tilde n_{\mathrm{PR}}^{(\mathrm{d})}\right]+\bar\rho\tilde w_{\mathrm{PR}}^{(\mathrm{d})}, \tag{10}$$

where turbulent transport is modeled according to an eddy diffusivity assumption. Assuming statistical independence between the progress variable c, the mixture fraction f, and the mixture fraction dissipation rate $\epsilon_{\mathrm{f}}=D\mid\nabla f\mid^2$, the Favre average of $w^{(\mathrm{d})}$, which is different from zero only in the region of combustion products, is given by

$$\bar\rho\tilde w^{(\mathrm{d})}=\overline{\rho w^{(\mathrm{d})}} =\mathrm{Prob}[c=1]\bar\epsilon_{\mathrm{f}}\int_0^1\rho_{\mathrm{b}}(\xi)P(\xi)\frac{\mathrm{d}^2 n_{\mathrm{PR}}^{\star}}{\mathrm{d}\xi^2}\,\mathrm{d}\xi, \tag{11}$$

where ξ represents the fuel mixture fraction in the scalar space and $\mathrm{Prob}[c=1]$ the probability of finding products. In case of flamelet combustion, where the local probability of finding $0<c<1$ is negligible, we have $\bar c=\mathrm{Prob}[c=0]\times 0+\mathrm{Prob}[c=1]\times 1=\mathrm{Prob}[c=1]$. It can also be shown that $\bar\rho_{\mathrm{b}}\bar c=\bar\rho\tilde c$. Introducing the Favre probability density function for the mixture fraction $\tilde P(f)=\rho_{\mathrm{b}}(f)P(f)/\bar\rho_{\mathrm{b}}$, Eq. (11) becomes

$$\bar\rho\tilde w^{(\mathrm{d})}=\bar\rho\tilde c\bar\epsilon_{\mathrm{f}}\int_0^1\tilde P(\xi)\frac{\mathrm{d}^2 n_{\mathrm{PR}}^{\star}}{\mathrm{d}\xi^2}\,\mathrm{d}\xi=\bar\rho\tilde c\bar\epsilon_{\mathrm{f}}\widetilde{\frac{\mathrm{d}^2 n_{\mathrm{PR}}^{\star}}{\mathrm{d}f^2}}. \tag{12}$$

The scalar dissipation rate is modeled here as $\bar\epsilon_{\mathrm{f}}=\widetilde{f''^2}/\tau_{\mathrm{m}}$, where τ_{m} is the turbulent mixing time. In a Reynolds-averaged Navier–Stokes (RANS) approach based on k–ϵ transport, this is given by $\tau_{\mathrm{m}}\simeq k/\epsilon$. The average concentration of prompt NO_x in the hot gas volume is therefore given by $\tilde n_{\mathrm{PR}}=\tilde n_{\mathrm{PR}}^{(\mathrm{ref})}+\tilde n^{(\mathrm{d})}$ with

$$\tilde n_{\mathrm{PR}}^{(\mathrm{ref})}=\tilde c\int_0^1 n_{\mathrm{PR}}^{\star}(\xi)\tilde P(\xi)\,\mathrm{d}\xi. \tag{13}$$

As shown in Fig. 1, postflame NO_x under lean fully premixed conditions is formed at a constant rate, hence still far from equilibrium conditions, where forward formation reactions are balanced by the backward ones. The following transport equation is therefore used here to account for postflame NO_x formation,

$$\bar\rho\frac{\partial\tilde n_{\mathrm{PF}}}{\partial t}+\bar\rho\tilde{\mathbf{u}}\nabla\tilde n_{\mathrm{PF}}=\nabla\cdot(\bar\rho D_{\mathrm{t}}\nabla\tilde n_{\mathrm{PF}})+\bar\rho\tilde w_{\mathrm{PF}},$$
$$\tilde w_{\mathrm{PF}}=\tilde c\int w_{\mathrm{PF}}^{\star}(\xi)\tilde P(\xi)\,\mathrm{d}\xi, \tag{14}$$

where $w_{\mathrm{PF}}^{\star}(f)$ is the constant postflame NO_x production rate function of the fuel mixture fraction f obtained in the postflame region from the calculation of 1D freely propagating flames with CHEMKIN II–PREMIX. This model has the advantage, over the more usual Zeldovich closure, of accounting also for additional significant postflame NO_x contributions such as the one from the $\mathrm{N_2O}$ pathway. The probability density function $\tilde P(f)$ is assumed here with the shape of the beta-function and parameterized with

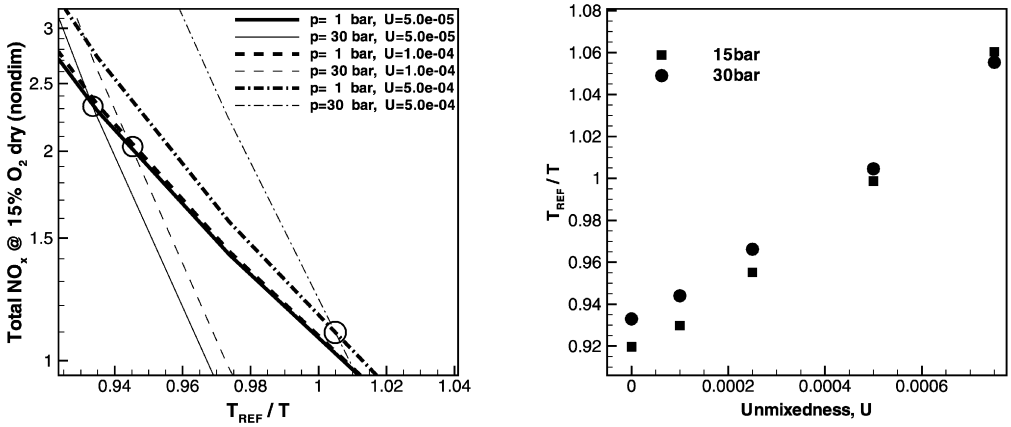

Fig. 8. Left: NO_x versus T_{REF}/T at 1 and 30 bar at different levels of unmixedness. Right: temperature where NO_x concentrations are equal to 1-bar NO_x levels versus fuel–air unmixedness (NO_x estimated over 5 ms).

respect to the mean $\tilde{f}$ and variance $\widetilde{f''^2}$ of the fuel mixture fraction [21].

The sources for the prompt NO_x correction and postflame NO_x, respectively in Eqs. (10) and (14), are subject, within the mean flame brush, to modulation by the mean reaction progress variable $\tilde{c}$. The distribution of the mean progress variable $\tilde{c}$, which defines the shape and position of the mean turbulent flame, depends of the turbulent burning rate U_t (see for example [22] for a detailed description of current state-of-the-art modeling of U_t) and on the convective and turbulent inflow of reactants into the mean flame. The flame, in fact, once anchored at a fixed location via some sort of piloting (e.g., recirculation of products), assumes a shape that ensures a local inflow of reactants equal to the turbulent burning rate. This condition is expressed analytically by the transport equation for the mean progress variable based on a turbulent burning rate closure [23],

$$\bar{\rho}\frac{\partial \tilde{c}}{\partial t} + \bar{\rho}\tilde{\mathbf{u}} \cdot \nabla \tilde{c} = \nabla \cdot (\bar{\rho} D_t \nabla \tilde{c}) + \rho_u U_t |\nabla \tilde{c}|. \tag{15}$$

The turbulent burning rate U_t is modeled using the turbulent flame-speed closure (TFC) model developed in [11],

$$U_t = A u' \mathrm{Da}^{1/4}, \tag{16}$$

with $\mathrm{Da} = \tau_m/\tau_{HR}$ and $\tau_{HR} = \lambda/s_L^2$. In this expression A is a proportionality constant whose optimal value of 0.5 has been found in the past [24] by comparing predictions obtained with the standard k–ϵ model with turbulent burning rate experimental data in a stirred spherical vessel.

3.2. Combined effect of unmixedness and operating pressure: one-dimensional analysis

The model given by Eqs. (10) and (14) is applied in a simple one-dimensional form in order to make a preliminary assessment of the effect of unmixedness and pressure on NO_x emissions. In gas turbine combustors, fuel–air unmixedness decreases rapidly on moving from inside the burner toward the combustor. Average hot gas temperatures are also subject to a fast drop in the combustor due to heat losses and mixing with liner cooling air. For these reasons NO_x production rates are significant only in the burner and in a small hot core in front of it. In this simple analysis, it is assumed that fuel–air unmixedness decays over a characteristic time τ_m that is comparable to the residence time in the hot core just in front of the exit of the burner. In the case of the EV burner variant considered here, this time is on the order of 5–10 ms. The mean flame is assumed to be narrow and flat so that 1D equations solved in the postflame region (following the fluid particles, i.e., according to a Lagrangian approach) are

$$\frac{D\tilde{n}_{PF}}{Dt} = \tilde{w}_{PF}, \quad \frac{D\tilde{n}_{PR}^{(d)}}{Dt} = \frac{\widetilde{f''^2}}{\tau_m}\widetilde{\frac{d^2 n_{PR}^{\star}}{df^2}},$$
$$\frac{D\widetilde{f''^2}}{Dt} = -\frac{\widetilde{f''^2}}{\tau_m}. \tag{17}$$

The mean sources are calculated according to relations (12) and (14) with the beta-shaped probability density function $\tilde{P}(f)$.

Fig. 8 left shows the total NO_x emissions calculated for a residence time of 5 ms vs T_{REF}/T at pressures of 1 and 30 bar for different initial values of the fuel–air unmixedness U, defined as $U = \widetilde{f''^2}/\tilde{f}/(1-\tilde{f})$. Fig. 8 right shows instead the temperature at which 20- and 30-bar NO_x are equal to

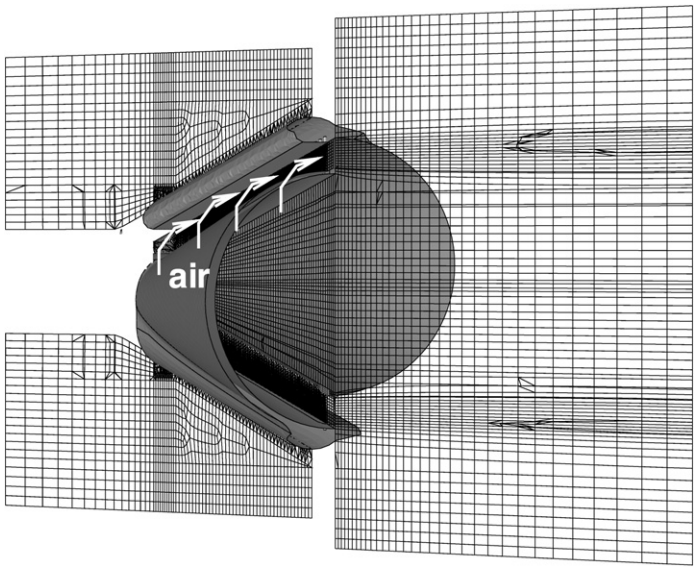

Fig. 9. Sketch of the ALSTOM EV burner and mesh in a longitudinal section used for the CFD calculations.

1-bar NO_x plotted versus unmixedness. In line with the discussion in the previous section, the characteristic flame temperature at which high-pressure and atmospheric NO_x become equal decreases with increasing unmixedness. This is a consequence of the fact that atmospheric NO_x is mostly of prompt type and therefore much less sensitive to unmixedness than 30-bar NO_x, which is mostly of postflame type. The increase of atmospheric NO_x with unmixedness is therefore considerably lower than the increase of high-pressure NO_x, a situation that leads to the shift of the NO_x intersection temperature (indicated with circles in Fig. 8, left) to lower value of adiabatic flame temperature.

The simple analysis performed here gives the opportunity to justify on a qualitative basis different NO_x pressure exponents reported in the literature. A detailed evaluation of the NO_x emissions from the experimental devices used in those investigations is outside the scope of the present work. The present work, however, suggests that the difference in the measured NO_x pressure exponents is due to the particular combination of residence times, equivalence ratio, and unmixedness effects characterizing those devices. For example, the experiments performed in a jet-stirred perfectly premixed reactor cavity over residence times up to 4 ms in [3,4] gave a significant reduction of NO_x emissions with pressure, in line with the present simple analysis under the same conditions. Higher levels of unmixedness and/or residence times might have instead characterized those experiments where a positive pressure exponent was detected (e.g., [8]).

4. CFD analysis of NO_x emissions in the ALSTOM EV double cone burner

Industrial dry low-NO_x burners usually stabilize the flame at a central recirculation zone (CRZ) obtained via the phenomenon of vortex breakdown. This phenomenon is briefly explained here for the case of the ALSTOM EnVironmental (EV) double cone burner sketched in Fig. 9. This burner, which has been the subject of previous experimental [17] and computational investigations [18–20], is obtained by shifting two halves of a cone in opposite directions, orthogonal to the cone axis, so that two inlet slots of constant width are produced. The air stream, when flowing through the burner slots, is given a strong swirl component. At the same time, fuel is injected through holes uniformly distributed along the burner slots, orthogonal to the stream of air. This ensures that all air streamlines through the slots are equally fed with fuel; i.e., it ensures good (but not perfect) fuel–air premixing prior to combustion. When the swirling flow reaches the expansion area between burner and combustor, the fluid particles radially diverge and slow down in the azimuthal direction due to the constraint imposed by the conservation of circulation $\Gamma = wr$. The vorticity vector $\boldsymbol{\omega}$ rotates with the streamlines in the azimuthal direction so that part of the axial vorticity is converted to azimuthal vorticity, which, according to the Biot–Savart law, induces a negative axial velocity at the axis [25]. If the negative axial velocity induced at the axis is sufficiently large, breakdown of the axial vortex takes place and a CRZ at the axis is created. The flame anchors at the axis, in proximity

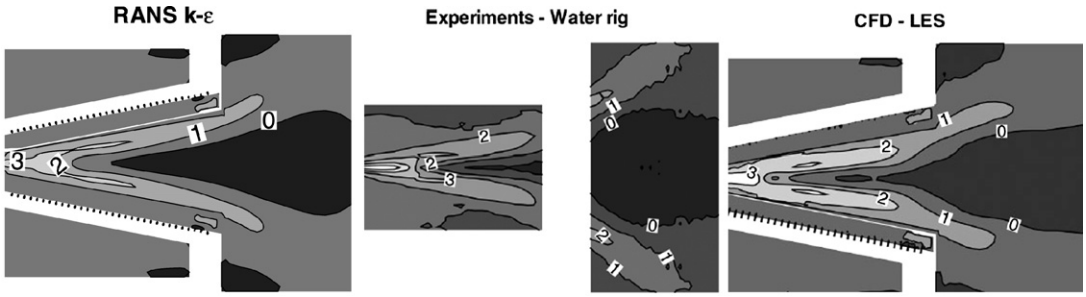

Fig. 10. Mean axial velocity in a longitudinal cross section of the EV burner. Nonreacting conditions. Left: RANS k–ϵ model. Center: experimental data. Right: LES model.

to the front stagnation point of the CRZ, where axial convective velocities are sufficiently low.

From a pure NO_x emissions point of view, the optimal situation is that reactants reach the flame completely premixed. However, during the burner development process, fuel–air unmixedness at the flame is used as an additional degree of freedom that helps to provide good flame-anchoring properties, i.e., sufficiently high margin from lean blow out and low-pressure pulsations due to thermoacoustic combustion instabilities (these are notoriously related to flame position by fuel injection–flame time lag effects). Fuel injection is therefore optimized for emissions and flame stability simultaneously; i.e., industrial "premixed" burners are in general always characterized by some residual fuel–air unmixedness at the flame.

The CFD model for prompt and postflame NO_x presented in the previous section is applied here to the ALSTOM EV double-cone burner. All CFD calculations have been performed using a commercial unstructured CFD finite-volumes solver (FLUENT version 6.2.16). All geometric quantities and velocities are nondimensionalized with the burner diameter D and burner bulk velocity U_0. The combustor has a circular cross section with nondimensional diameter and length respectively equal to 2 and 7. To improve numerical accuracy and overcome difficulties connected with the complexity of the geometry, the computational domain has been meshed with a mixed tetrahedral–hexahedral mesh of approx. 400,000 elements, with the hexahedrals concentrated in the central region of the burner and in the combustor. The finite-volume elements inside the burner and in the combustor close to the burner exit have an average nondimensional size of 0.025. A longitudinal cross section of the burner–combustor domain showing the computational mesh is presented in Fig. 9. The RANS calculations have been run with the realizable version [26] of the k–ϵ transport model and with numerics based on the QUICK scheme [27]. The LES simulations are based instead on the standard Smagorinsky subgrid model, with central differencing used for all variables with the exception of the transport equation for the fuel mixture fraction, which is discretized with the QUICK scheme [27]. All LES simulations have been run using second-order discretization in time and a nondimensional time step $t/(D/U_0) = 0.014$. The calculations have been performed on four nodes of a LINUX cluster. The LES calculations have first been run for a time sufficiently long to yield a statistically steady solution. Final averaging has then been performed over a total time $t/(D/U_0) \simeq 10$, i.e., approx 700 time steps, which required around 12 h of CPU time for each case.

The combusting flow is calculated using the TFC combustion model according to its standard formulation based on Eq. (15). In the case of the LES calculations, the model constant A has been taken equal to 0.9, higher than the standard RANS value of 0.5. A discussion explaining the reasons for this higher value of the constant in LES simulations is given in [20]. Previous work [20] has shown that RANS methods are not suitable for correctly reproducing the flame anchoring properties of this burner. It was in fact found that RANS methods based on the k–ϵ transport model do not correctly reproduce the characteristics of the CRZ produced by the vortex breakdown phenomenon, which are very important in determining the anchoring location of the turbulent premixed flame at the axis. Accurate results have instead been obtained via LES. This issue is documented in Fig. 10, which shows a comparison of the mean axial velocity flowfield obtained from the RANS and LES calculations with experimental data from a water rig facility. While the position of the apex of the CRZ is correctly predicted with the realizable k–ϵ method (deep in the burner), the radial size is not. Much better agreement with experiments, in terms of both the position and the radial size of the CRZ (see Fig. 10, right), is instead obtained via LES.

It has been shown in [20] that the radial size of the CRZ has a strong effect on the anchoring position at the axis of the turbulent premixed flame. The turbulent premixed flame stabilized at a CRZ, in fact,

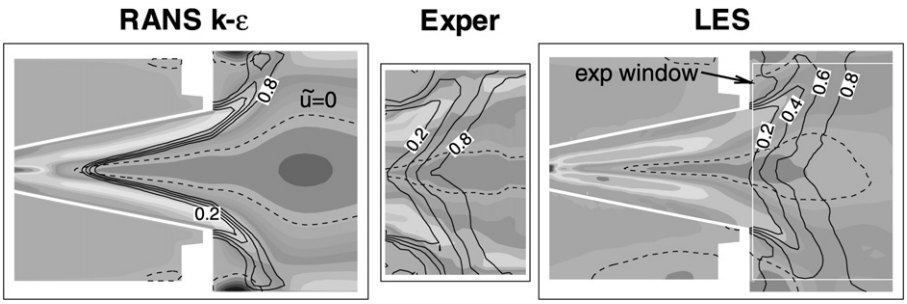

Fig. 11. Mean axial velocity (contours) and mean progress variable (isolines) in a longitudinal cross section of the EV burner at $T = T_{\mathrm{REF}}$. Fully premixed conditions. Left: RANS k–ϵ model. Center: experiments. Right: LES model.

is not flat like the one stabilized in a flow impinging against a flat wall but, as shown in [20], it is characterized by a curvature radius R_{F} at the axis, which is inversely proportionally to the radial variation of the mean axial velocity, this last measured via the factor $\mathrm{d}^2\tilde{u}/\mathrm{d}r^2$. In this case, the turbulent flame does not receive reactants at the axis only in the form of mean axial convective flux but also as a radial turbulent flux, which in cylindrical coordinates can be expressed as $-2(D_{\mathrm{t}}/R_{\mathrm{F}})\,\mathrm{d}\tilde{c}/\mathrm{d}r$. Assuming a velocity flowfield like the experimental nonreacting one, a flame placed inside the burner would have very small curvature radius R_{F} due to the very narrow CRZ (high $\mathrm{d}^2\tilde{u}/\mathrm{d}r^2$) and therefore would be subject to a very high radial turbulent inflow of reactants, which was estimated in [20] to be much larger than typical values of the turbulent burning rate. Using a simple 1D method, it has been shown in [20] that, in this case, the balance between the inflow of reactants to the flame and the turbulent burning rate, takes place at a position inside the CRZ close to the burner exit where the CRZ starts increasing in thickness in downstream direction and therefore the estimated R_{F} is not too small; i.e., the flame anchors in a intermediate position, completely intersecting the CRZ. Using the standard definition of turbulent flame speed as the component of the flow velocity orthogonal to the flame, this means that the flame in the EV burner has negative speed at the axis. In case of RANS methods, due to the significant overestimation of radial size of the CRZ (see Fig. 10, left), the radial flux of reactants into a flame placed inside the burner would be much less than in the case of LES and therefore the flame mass balance is satisfied much deeper in the burner. These estimates on the flame position are confirmed by the combusting CFD calculations under fully premixed conditions at $T = T_{\mathrm{REF}}$ shown, together with the experimental data, in Fig. 11. The RANS method gives in fact a flame that anchors deeply inside the burner, while LES gives a flame that anchors in the combustor, intersecting the CRZ just downstream of the mixing tube exit, in line with the experimental data. Given these results, it was decided here to perform the NO_x emission analysis with the LES method.

In order to limit the CPU time, postflame and prompt NO_x emissions have not been determined via the solution of their transport equations but instead via the volumetric integration of their production rates. In fact, integrating Eqs. (12) and (14) over the computational domain yields the net flux (kmol/s) of prompt NO_x correction and postflame NO_x as

$$\mathrm{Flux}_{\mathrm{PR}} = \int_{\mathrm{Exit}} \bar{\rho}\tilde{u}\tilde{n}_{\mathrm{PR}}^{(\mathrm{d})}\,\mathrm{d}A = \int_{V} \bar{\rho}\tilde{w}_{\mathrm{PR}}^{(\mathrm{d})}\,\mathrm{d}V, \tag{18}$$

$$\mathrm{Flux}_{\mathrm{PF}} = \int_{\mathrm{Exit}} \bar{\rho}\tilde{u}\tilde{n}_{\mathrm{PF}}\,\mathrm{d}A = \int_{V} \bar{\rho}\tilde{w}_{\mathrm{PF}}\,\mathrm{d}V. \tag{19}$$

The total prompt NO_x emissions at the combustor exit are obtained by adding to the prompt NO_x correction estimated from relation (18), the prompt NO_x concentration calculated with CHEMKIN–PREMIX at the bulk equivalence ratio in the combustor. NO_x experimental data are available at 1 and 15 bar for variations of the bulk adiabatic temperature and a fixed value of the burner bulk velocity U_0 (i.e., fixed residence time).

NO_x emissions obtained from the LES simulations are compared with experiments in Fig. 12 as functions of the adiabatic temperature of the combustion products. Fig. 12 left shows that the calculated NO_x emissions underestimate the experimental values and, not surprisingly, the error is higher at 15 bar then at 1 bar. Given in fact the higher sensitivity to temperature of high-pressure NO_x emissions, errors in the estimation of fuel–air unmixedness have a larger impact on NO_x at 15 bar than at 1 bar. The calculated NO_x emissions show, however, a pressure exponent that increases with T_{REF}/T from $\simeq$0.1 to $\simeq$0.67, consistent with the experimental data, where the NO_x pressure exponent increases from approximately 0.1 to approximately 0.76. Both the numerical and experimental data do not show any intersection between the 15- and 1-bar NO_x curves (no temperature range where the pressure exponent is below

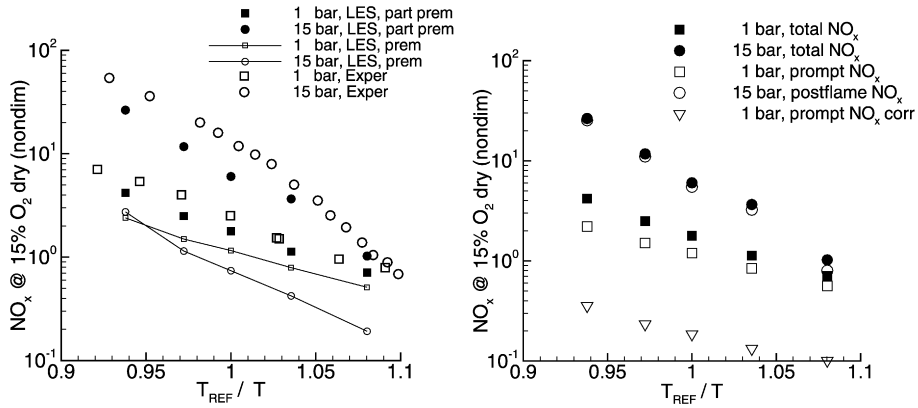

Fig. 12. Comparison of NO_x emissions from LES with experiments in the EV burner. Left: total NO_x. Right: split into postflame and prompt NO_x contributions.

zero). In line with the 1D results, the right part of Fig. 12 shows that in the 15-bar case, most of the NO_x is of postflame type (especially in the high-temperature range), while in the 1-bar case, most of the NO_x is of prompt type (especially in the low-temperature range, where postflame NO_x is negligible at all pressures). Finally, the prompt NO_x correction $\tilde{n}_{PR}^{(d)}$ at 1-bar accounts for approx 10% of the total prompt NO_x contribution.

LES has also been run under fully premixed conditions, in order to evaluate the theoretical NO_x potential of the burner. The results, shown in Fig. 12 left for this case, indicate that, in the absence of fuel–air unmixedness and in line with the 1D analysis, high-pressure NO_x is lower than atmospheric NO_x over a rather large range of flame temperature. Such low NO_x emissions are, however, not easy to achieve in an industrial combustor due to the fact that a given level of fuel–air unmixedness is usually necessary to ensure flame stability. In other words, the problem of achieving single-digit NO_x emissions from industrial burners is a pure flame stability one.

5. Conclusions

The effect of pressure and fuel–air unmixedness on NO_x emissions from industrial gas turbines burners has been analyzed here using results from the CHEMKIN II premix module with GRI3.0 mechanisms and a model where the total NO_x formation is decomposed, according to a Damköhler number criterion, to a fast "prompt" contribution produced within the heat release region and a slow "postflame" one. The analysis shows that (a) the flamelet "prompt" contribution is around a factor of 3 less sensitive to fuel–air unmixedness than the "postflame" one, (b) the sensitivity to unmixedness of "prompt" and "postflame" NO_x does not strongly depends on pressure, and (c) the "prompt" and "postflame" contributions change with pressure respectively with exponents $\simeq -0.45$ and $\simeq 0.67$.

A preliminary analysis has been carried out using a simple 1D approach and a model for NO_x emissions developed here that separately accounts for the prompt and postflame contributions. Based on this analysis, the combination of the NO_x emissions properties from CHEMKIN II–GRI3.0 chemistry, under fully premixed conditions and typical hot gas residence times in industrial burners, shows that high-pressure (e.g., 20-bar) NO_x emissions can in principle be lower than atmospheric ones (negative pressure exponent of total NO_x). Given, however, that postflame NO_x represents the dominant contribution at high-pressure and given its highest sensitivity to flame temperature variations, in the presence of even a small amount of fuel–air unmixedness or high residence times, the situation can easily be reversed in favor of atmospheric NO_x emissions. The problem of reducing NO_x emissions to single-digit values in industrial gas turbines is therefore a problem of sorting out those flame stability issues that imply keeping some degree of fuel–air unmixedness at the flame.

The results from the simple 1D analysis are finally confirmed by LES simulations of the combusting flow in the ALSTOM industrial EV burner. These, in line with experiments, give a positive NO_x pressure exponent increasing with equivalence ratio from $\simeq 0.1$ to $\simeq 0.67$ under standard fuel injection, but a negative pressure exponent could be obtained over a large range of the equivalence ratio under ideal fully premixed conditions.

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References

[1] H. Kobayashi, T. Tamura, T. Takashi, F. Williams, Proc. Combust. Inst. 26 (1996) 389–396.
[2] D.A. Sullivan, J. Eng. Power Trans. ASME 99 (1977) 145–152.
[3] K.U.M. Bengtsson, P. Benz, R. Schaeren, C.E. Frouzakis, Proc. Combust. Inst. 27 (1998) 1393–1401.
[4] T. Rutar, P.C. Malte, in: Proc. of the ASME: TURBO EXPO 2001, June 4–7, 2001, New Orleans, LA, USA.
[5] R.C. Steele, J.H. Tonouchi, D.G. Nicol, D.C. Horning, P.C. Malte, D.T. Pratt, J. Eng. Gas Turbines Power Trans. ASME 120 (1998) 303–310.
[6] S.M. Correa, Combust. Sci. Technol. 87 (1992) 329–362.
[7] G. Leonard, J. Stegmaier, ASME Paper 93-GT-288 (1993), ASME, New York.
[8] R.K. Mongia, E. Tomita, F.K. Hsu, L. Talbot, R.W. Dibble, Proc. Combust. Inst. 26 (1996) 2749–2755.
[9] T.F. Fric, J. Propulsion Power 9 (1993) 708–713.
[10] J.C. Barnes, A.M. Mellor, J. Propulsion Power 14 (1998) 974–980.
[11] V.L. Zimont, Combust. Explos. Shock Waves 15 (1979) 305–311.
[12] R.J. Kee, F.M. Rumpley, J.A. Miller, CHEMKIN-II: A Fortran Chemical Kinetics Package for the Analysis of Gas Phase Chemical Kinetics Gov. Pub. SAND89-9009B, September, 1989.
[13] Available at http://www.me.berkeley.edu/gri_mech/, accessed 28 May 2007.
[14] R. Carroni, T. Griffin, J. Mantzaras, M. Reinke, Catal. Today 83 (2003) 157–170.
[15] D.G. Nicol, R.C. Steele, N.M. Marinov, P.C. Malte, J. Eng. Gas Turbines Power Trans. ASME 117 (1995) 100–111.
[16] N. Peters, J. Fluid Mech. 384 (1999) 107–132.
[17] F. Dinkelacker, A. Soika, D. Most, D. Hofmann, A. Leipertz, W. Polifke, K. Döbbeling, Proc. Combust. Inst. 27 (1971) 857–865.
[18] V.L. Zimont, W. Polifke, M. Bettelini, W. Weisenstein, J. Eng. Gas Turbines Power Trans. ASME 120 (1992) 526–532.
[19] P. Flohr, in: R. Friedrich, B. Geurts, O. Métais (Eds.), Proc. of the 5th Int. ERCOFTAC Workshop on Direct and Large-Eddy Simulation, Munich University of Technology, 2003.
[20] F. Biagioli, Combust. Theory Modeling 10 (3) (2006) 389–412.
[21] V. Eswaran, S.B. Pope, Phys. Fluids 31 (1988) 506.
[22] A.N. Lipatnikov, J. Chomiak, Prog. Energy Combust. Sci. 28 (2002) 1–74.
[23] V.L. Zimont, Exp. Therm. Fluid Sci. 21 (2000) 179–186.
[24] V.L. Zimont, A.N. Lipatnikov, Chem. Phys. Reports 14 (7) (1995) 993–1025.
[25] G.K. Batchelor, An Introduction to Fluid Dynamics, Cambridge Univ. Press, 1967.
[26] T.-H. Shih, W.W. Liou, A. Shabbir, Z. Yang, J. Zhu, Comp. Fluids 3 (1995) 227–238.
[27] P. Leonard, Comput. Methods Appl. Mech. Eng. 19 (1979) 58–98.